

Marvis Osazuwa

Machine Learning Engineer · Data Scientist

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PROFILE

Machine Learning Engineer and Data Scientist focused on model serving and explainability: survival analysis, gradient-boosted classification with SHAP, in-database ML via PostgresML, served through FastAPI with JWT and Row-Level Security, and exposed to agentic LLM clients via a Model Context Protocol (MCP) endpoint. Strong on the modelling-to-serving handoff; CI-gated correctness through dbt and pytest. Most recent work exposed the intelligence layer to agentic LLM workflows via a Model Context Protocol (MCP) endpoint. Career spans retail banking (Nigeria, 2015 to 2016), mortgage analytics (Nigeria, 2018 to 2019), healthcare marketing (Türkiye, 2024 to 2025), and a most recent Analytics Engineer role at Force24 (UK, 2026); MSc Big Data Analytics (Bahçeşehir, 2020 to 2023) and MSc Data Science (Salford, 2025 to May 2026). Currently building independent enterprise projects in the agentic and analytics space, deepening on AI engineering via certifications. UK Graduate visa: full right to work in the UK, no sponsorship needed; EU Blue Card eligible for Germany and the EU.

EXPERIENCE

Analytics Engineer · Force24 · Leeds, UK (Hybrid)

Fixed-term contract · January 2026 – April 2026

Marketing automation SaaS. A 16-week collaborative build of a 3-tier customer intelligence platform. Within the team delivery I owned the data layer outright as sole architect and author; was primary author across 8 backend domains, and shipped the customer-facing frontend.

- Sole architect and author of a dbt and Dagster data pipeline that consolidated 4 disconnected business systems into one analytics-ready warehouse on a deduplicated account spine (over 40 dbt models, 5 daily schedules, ~93% solo commit share). Established a single canonical customer ID across the business, which closed the record-matching gaps that had made Customer Success books unjoinable.
- Cut daily pipeline compute by ~95% by re-architecting high-volume revenue models as incremental on an append-only raw layer with deterministic hash IDs. Productionised the platform on Oracle Cloud with idempotent bootstrap, TLS, VPN allowlist, and three-level operator runbooks, so it runs 24/7 at zero infrastructure cost.
- Made support attribution auditor-defensible with a 10-step deterministic classification chain that maps every ticket to exactly one account. This replaced a black-box best-guess join and closed the last open finding in the Customer Success metrics review.
- Gated the build behind 123 dbt tests and 82 pytest tests in GitHub Actions CI, which caught 22 data-quality bugs during the first production run. Silent metric drift now fails a test instead of surfacing in the boardroom.
- Shipped the customer-facing backend and Angular UI (customer list with 10 server-side filters, dashboard and reporting endpoints, sole-author Action Tracker across 7 outcome types), which gave Customer Success a single workflow for retention actions and produced the first labelled outcomes dataset for a planned causal uplift model.

Stack: Dagster, dbt, PostgreSQL, Python, FastAPI, Redis, Angular, Docker, Caddy, Oracle Cloud, GitHub Actions, Git.

CRM Data Specialist · Natural Clinic · Istanbul, Türkiye

June 2024 – January 2025

Healthcare marketing organisation. CRM data and analytics ownership across patient acquisition, conversion, and retention.

- Rebuilt the CRM data layer end to end as the single source of truth for sales, marketing, and clinical operations. Lead response time dropped 30%; conversion rose 15%; duplicates and data errors fell 40%; email engagement rose 25%.
- Replaced weekly manual spreadsheet reporting with real-time Power BI dashboards and exec-ready reports, cutting leadership reporting latency from 7 days to live and freeing ~6 analyst-hours per week on report assembly.
- Integrated the CRM with the marketing automation stack (email, SMS, paid funnels) and engineered customer data flows between CRM, patient management, and marketing tooling, which removed three days a week of manual reconciliation.
- Translated ambiguous stakeholder requests from marketing, clinical operations, and customer service into 12+ shipped data models and dashboards, including the lead-scoring and conversion-funnel models that drove the 15% conversion lift.

Stack: SQL, Python, Power BI, Zoho CRM, marketing automation, workflow automation.

Strategic Data Insights Analyst · Federal Mortgage Bank of Nigeria · Abuja, Nigeria

December 2018 – November 2019

Federal mortgage institution managing Nigeria's national housing fund. Loan portfolio analytics and IT-governance reporting in the Asset Creation Unit of the Loan Administration group.

- Contributed to a data-integrity overhaul that improved core dataset accuracy by 40% in six months, materially reducing the risk of faulty loan approvals.
- Built Python and SQL analytics on top of Tableau to track loan repayment behaviour, supporting a 15% lift in repayment-monitoring efficiency and a 25% reduction in manual data entry.
- Cut IT-asset-governance approval turnaround by 30% by re-mapping the approval workflow and instrumenting stage-level reporting in Tableau, shrinking median time-to-approve across the Loan Administration group.

Stack: Python, SQL, Tableau, Excel.

Customer Information and Service Specialist · Guaranty Trust Bank · Edo, Nigeria

September 2015 – March 2016

Early-career customer service role at a Tier-1 African retail bank.

- Supported the migration of in-branch customers to USSD and Internet banking through targeted onboarding sessions, owned complaint-resolution logging and root-cause categorisation for the branch monthly service-quality report, and trained incoming customer-service interns on KYC and complaint handling.

EDUCATION

MSc Data Science · University of Salford · Manchester, UK

January 2025 – May 2026 · Distinction · Dissertation: Agentic ELT Data Platform for Customer Intelligence

MSc Big Data Analytics and Management · Bahcesehir University · Istanbul, Türkiye

September 2020 – March 2023 · GPA 3.67 / 4.00 · Thesis: distributed PySpark pipeline for football scouting

BEng Computer Engineering · University of Benin · Edo, Nigeria

September 2011 – August 2017

PROJECTS

Agentic ELT Data Platform for Customer Intelligence

MSc Dissertation (Salford) · 2026 · Live B2B SaaS Environment

Solo research build, end to end: every layer designed and written by me alone, from ingestion to the serving API, against a live B2B SaaS environment under NDA. Ran on the same environment as the Force24 contract but is independent work; the Force24 platform itself was a team delivery.

- **Ingestion and warehouse:** custom Python extractors ingested over 1M records from multiple vendor APIs into a JSONB-first PostgreSQL warehouse, orchestrated with Dagster.
- **Serving and agent access:** FastAPI service with JWT auth and Row-Level Security, Angular dashboard, and a Model Context Protocol (MCP) endpoint queryable by any MCP-compatible LLM agent for per-account risk profiling, driver explanations, and intervention recommendations. Output prioritisation surfaced high-risk accounts to CSMs for retention action.
- **Modelling:** 48 dbt models across raw, staging, intermediate, and marts; layered Kimball-style dimensional design.
- **Intelligence stack:** survival analysis (cross-validated C-index ~0.94 on held-out cohort) for time-to-churn; gradient-boosted classification with SHAP explanations (AUC ~0.95 vs ~0.89 logistic baseline) for churn likelihood and drivers; DR-Learner causal inference (econml) correcting reactive-assignment selection bias by ~50 percentage points on the naïve estimator for honest CSM treatment effect.

Python PostgreSQL JSONB dbt Dagster PostgresML XGBoost SHAP Causal Inference Survival Analysis FastAPI Angular MCP

Pharmaceutical Side Effect Classification

Side Project · 2025 · Healthcare ML

Production-shape multi-class classifier mapping free-text adverse-event descriptions to ten clinical categories across 11,825 marketed medicines. Random Forest at 98.5% accuracy on held-out test, Logistic Regression baseline at 97.2% (interpretable comparison within two points). Above 95% F1 across every retained category. Inference CLI and pytest test suite shipped.

Python scikit-learn TF-IDF Random Forest Logistic Regression pytest

Conflict Sentiment Analysis

Side Project · 2026 · NLP / Media Analysis

Comparative methodology study quantifying how 8,158 English-language news articles framed the Russia and Ukraine conflict across 68 publishers and 18 countries. Ran three sentiment engines (RoBERTa, VADER, TextBlob) plus a gensim LDA topic model on the same corpus to surface where engine choice changes the answer. Solo build, end to end.

Python RoBERTa VADER TextBlob gensim LDA NLP Sentiment Analysis

Steam Games Recommender

Portfolio Project · 2023 · Distributed ML

Collaborative-filtering recommender on the Steam 200k implicit-feedback dataset. Distributed Alternating Least Squares training on Spark MLlib with hyperparameter sweeps via CrossValidator and TrainValidationSplit, full experiment tracking in MLflow, shipped on Databricks Community Edition. End-to-end ML lifecycle at scale.

PySpark Spark MLlib ALS MLflow Databricks Recommender Systems

TECHNICAL SKILLS

Languages: SQL, Python (pandas, NumPy, scikit-learn), R, PySpark, T-SQL, Bash
ML & Statistics: scikit-learn, XGBoost, PostgresML, scikit-survival, econml (DR-Learner), SHAP, MLflow, RoBERTa, ARIMA, Time Series Analysis, NLP, Sentiment Analysis, TF-IDF, LDA, Causal Inference, Survival Analysis, A/B testing / experimentation
Agentic AI: Model Context Protocol (MCP), agentic ELT, LLM-accessible data layers
Service Layer: FastAPI, REST APIs, Docker, Caddy, Model Context Protocol (MCP)
Warehousing & Storage: PostgreSQL (JSONB + GIN), Snowflake, BigQuery, Databricks, SQL Server
Transformation & Orchestration: dbt, Dagster, Apache Spark, Airflow
Source Control & CI: Git, GitHub, GitHub Actions, pytest
Cloud: AWS, GCP, Azure, Oracle Cloud
BI & Visualisation: Power BI, Tableau, Looker

LANGUAGES

- **English:** Native
- **German:** B1

CERTIFICATIONS

- Engineer Data for Predictive Modeling with BigQuery ML · Google Cloud
- Python for Data Science, AI & Development · Coursera
- Generative AI for Business Leaders · Coursera
- Enterprise Design Thinking Practitioner / Team Essentials for AI / Co-Creator · IBM

COMMUNITY & LEADERSHIP

Microsoft Learn Student Ambassador · Bahçeşehir University · Istanbul, Türkiye

January 2021 – September 2023 (2 yrs 9 mo) · Cross-university Azure workshops and a community hackathon with fellow ambassadors and interested learners.

Availability: Available immediately for UK roles (UK Graduate visa, no sponsorship needed) and for Germany and the EU (EU Blue Card eligible).